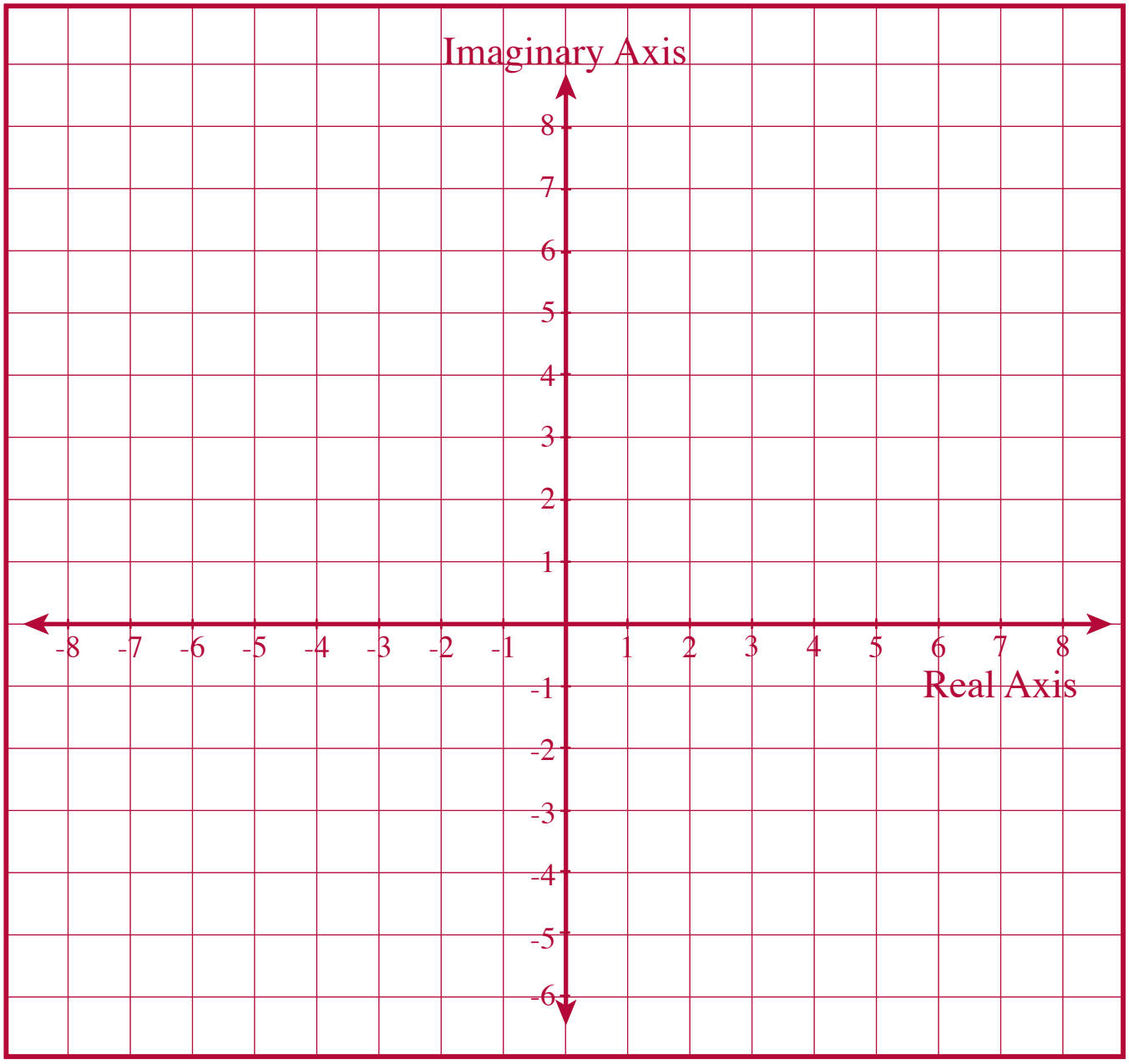
Student Activity 1

Addition and Subtraction of Complex Numbers

1. Add $z = 4 + i$ to each of the following complex numbers:
   o $= 0 + 0i$
   $w_1 = 2 + 2i$
   $w_2 = -3 + 2i$
   $w_3 = 0 + 4i$
2. Represent the complex numberso, w_1, w_2, w_3, as points on an Argand Diagram and then show the results from the above exercise using a directed line (a line with an arrow indicating direction) between each w and its corresponding $w + z$. What do you notice?



Student Activity 2

Multiplication by an Imaginary Number

1. If $z = 3 + 4i$, what is the value of iz, i^2z, i^3z, i^4z? Represent your results on an Argand Diagram joining each point to the origin o = 0 + 0i.
2. Investigate what is happening geometrically when z is multiplied by i to get iz? Use geometrical instruments and/or calculation to help you in your investigation.
3. Prove true for the multiplication of iz by i that you get i^2z and the multiplication of i^2z by i that you get i^3z etc.
4. Write your conclusion.
5. Plot on an Argand Diagram $4 + 2i$ and $-i$. Multiply $-i\,(4 + 2i)$. What do you notice?

© Project Maths Development Team 2010